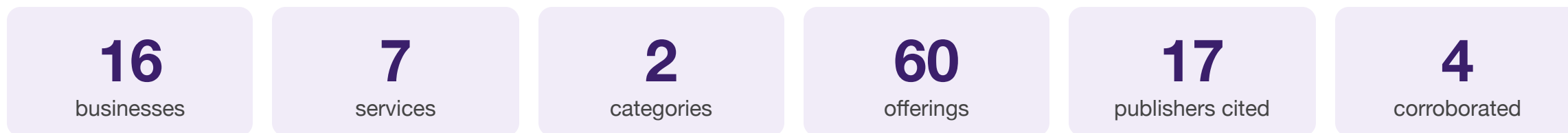


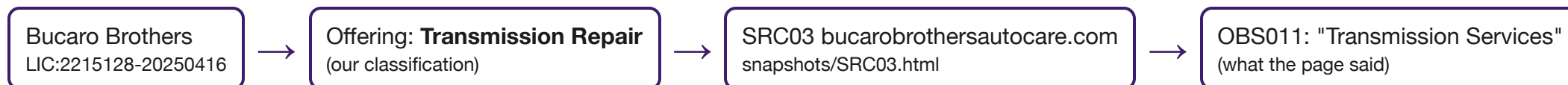
Chicago North Side Car Repair Directory

CS 284 / CS 430 Project 1 · Project 1 Team: Saron, Eyael, Tutu · MySQL 8.4.5 · snapshot 2026-09-30

- **User question:** which nearby repair shops advertise the service I need, and how do we know?
- **Scope:** shops with a City of Chicago *Motor Vehicle Services License* in Lake View, Lincoln Park, Lincoln Square, North Center



Source trace: observation vs. classification



Model, keys, and normalization

```
service_category (1) —< (N) service
service (1) —< (N) business_service (N) >— (1) business
source (1) —< (N) business_service
source (1) —< (N) business

business_service PK = (business_id, service_id)
```

- **1:N** category → service (FK NOT NULL)
- **M:N** business ↔ service via business_service
- Composite PK blocks duplicate offerings (BR4)
- Every business and offering cites a source + raw-row ref (BR3)
- Delete business → CASCADE offerings; others RESTRICT

From one collection table to 3NF

```
R(name, address, website, service, category,
  wording, obs_row, source_url, publisher, date)
key: {name, address, service}
FD1 name,address → website ... (partial)
FD2 service → category (partial)
FD3 source_url → publisher, date (transitive)
```

- Split into business · service · service_category · source · business_service
- **Anomaly removed:** Fred's website was repeated on 7 rows → now stored once
- FKs rejoin every original row (the Q2 join)

Build and validation

```
mysql -u root -p < sql/00_schema.sql      # reset + tables
mysql -u root -p < sql/01_load.sql        # real snapshot
mysql -u root -p < sql/02_views.sql       # directory view
mysql -u root -p < sql/03_queries.sql     # Q1-Q6
tests/run_tests.sh                       # T1-T4 log
```

- No network needed; FK checks ON
- Rebuilt twice → identical counts (16/7/2/60/21)

Test	Expected	Result
T1 counts + source links	all ≥ minimums, 0 broken links	PASS
T2 FK: business 999 / service 99	ERROR 1452 ×2	PASS
T3 duplicate pair (3,1)	ERROR 1062	PASS
T4 fixture: Q4 / Q5 / Q6 / Q3	901,902 · NULL rows · 901 · 3,2	PASS

T4 runs in the separate database p1_fixture, which is dropped afterwards. The real snapshot has 0 fixture rows.

Queries and findings

Q3: services with ≥ 7 providers

Service	Providers
Vehicle Diagnostics	14
Brake Repair	11
Tire Service	9
Oil Change	8
A/C & Heating Repair	7

Below threshold: Transmission Repair 6, Wheel Alignment 5

- **Q2:** 6 shops record Transmission Repair, each with a source URL + raw row (no duplicate providers: PK)
- **Finding (Q4):** 11 of 16 shops have *no alignment recorded*, including both Lincoln Square shops
- **Q5:** LEFT JOIN with the filter in ON keeps NULL rows (Mechanista: 4 matched, 3 NULL)
- **Limitation:** absence = *not recorded*, not "not offered"; mostly home-page evidence; 16-shop sample ≠ market demand
- **Follow-up:** do the 11 shops actually do alignments? Check their service pages or call.